

Daniel Sardak

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EDUCATION

PhD in Data Science, Worcester Polytechnic Institute Aug 2025 - May 2030

MS in Artificial Intelligence, Worcester Polytechnic Institute May 2025

BS in Computer Science, Worcester Polytechnic Institute Dec 2024

RESEARCH AND PUBLICATIONS

Contested logistics: Resilience of strategic highways and railways | *GIS, Network Science* July 2025

- **Publication:** Transportation Research Interdisciplinary Perspectives
- **Authors:** Sukhwan Chung, **Daniel Sardak**, Maksim Kitsak, Andrew Jin, Igor Linkov

Assessing the Robustness and Resilience of U.S. Strategic Highways: A Network Science Perspective | *Resilience, Networks*

- **Publication:** Transportation Research Record
- **Authors:** Sukhwan Chung, **Daniel Sardak**, Jeffrey Cegan, Igor Linkov

EXPERIENCE

Teaching Assistant | WPI Computer Science Department Aug 2025 - Present

- Led office hours to assist students with homework questions and concepts for WPI's Information Retrieval course
- Created grading rubrics and homework solutions for the course
- Designed and led a lecture on basic machine learning and deep learning using PyTorch

Research Intern | US Army Corps of Engineers June 2024 - Aug 2025

- Developed a dataset to assess the compounded effects of natural disasters and developed a predictive model analyzing how wildfires influence flood intensity.
- Applied network science methods to assess and quantify the resilience of road and rail infrastructures against natural disasters.
- Trained a image segmentation model with 80% accuracy to detect water towers from satellite images
- Developed a simulation and analysis framework to compare multiple disaster recovery strategies
- Developed an LLM-powered application to automatically generate environmental proposals

Peer Learning Assistant | WPI Math Department March 2022 - Aug 2025

- Led weekly discussion and tutoring sessions for Statistics, Calculus, and Linear Algebra courses for 30+ students
- Graded homework and exams, and provided timely feedback on student's understanding of the material

PROJECTS

RAG Powered Immigration Form Filler | RAG, LLM, Haystack Jan 2025 - May 2025

- Developed a RAG powered immigration form filler powered by Mistral to fill out immigration forms
- Used Haystack to develop an end-to-end RAG pipeline
- Featured a chatbot interface to upload documents and collect user information

Text-to-Image Model Geographic Bias Analysis | Text-to-Image, LLM, Bias Analysis Jan 2025 - May 2025

- Evaluated geographic bias in text-to-image models using automatic and human-annotated metrics
- Found systematic gap: human evaluators flagged stereotyping in underrepresented regions while automatic metrics rated images as higher quality

LLM Powered DBMS Assistant | LLM, SQL, Python Aug - Dec 2023

- Developed an LLM powered DBMS assistant that uses GPT-3.5-turbo to answer questions about a database
- Created a pipeline to convert a schema and user query into a SQL query
- Developed a chatbot interface to interact with the DBMS assistant

TECHNICAL SKILLS

Programming Languages: Python, Java, Bash, JavaScript, jQuery, CSS3, HTML5, R

Applications: Tableau, Data Analysis, Machine and Deep Learning, Data Visualization, Data Engineering, Web Development

Database/Warehousing: MySQL, Google Cloud, PostgreSQL, Azure, Hadoop, Spark